

STRATEGIC VENDOR INTELLIGENCE (SVI)

STRATEGIC OBJECTIVE: IMPROVE PROCUREMENT WITH 22-FEATURE ML SEGMENTATION

Executive Impact:

Transforming Data into Strategy: In a high-growth startup environment, traditional "Sales Sorting" is a liability. This project transitions the procurement function from 1-dimensional ABC sorting to a Multi-Dimensional Strategic Map. By analyzing 22 operational drivers, we identify exactly where efficiency is lost in the catalog and where supply volatility is hindering scaling efforts.

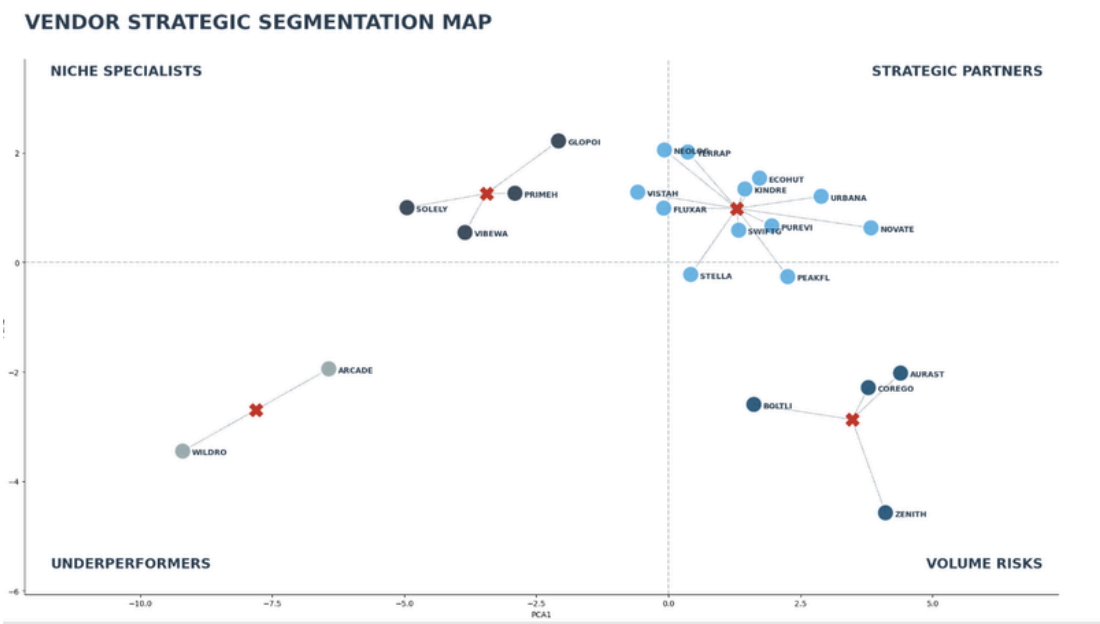
Problem Statement

Beyond the Pivot Table: Managing an expanding vendor portfolio via "gut feeling" ignores the hidden operational costs of scaling. High sales often mask catalog bloat (excessive C-items) and supply risks (high out-of-stock rates). The business required a system to turn standard, readily available data into a prescriptive action plan for future buying cycles.

Methodology

Plug-and-Play Intelligence: The framework is designed for immediate adoption. It requires only a standard ABC Vendor Export—a data format easily generated via basic Excel pivot tables already used by the buying team.

- Feature Engineering: Expanded basic sales data into 22 weighted metrics, including Portfolio Purity, Revenue Density, and Buffer Reliability.
- Dimensionality Reduction: Utilized PCA (Principal Component Analysis) to condense 22-dimensional performance into a scannable 2D Strategic Map.
- ML Clustering: Applied K-Means Clustering to segment the portfolio into 4 actionable archetypes: Strategic Partners, Niche Specialists, Volume Risks, and Underperformers.



Key Performance Outcomes

Measurable Operational Gains:

- 15% Reduction in Purchasing Costs: Optimized negotiation leverage and identified hidden operational overhead.
- 23% Sales Growth on 'A' Products: Re-prioritized logistics to ensure top-selling items reached maximum availability.
- 25% Reduction in C-Items: Systematically delisted dead-weight items, releasing significant working capital.

The Bottom Line Summary:

Transformed procurement from a reactive function (historical sales tracking) into a prescriptive profit engine. By engineered a 22-feature Machine Learning model, I converted standard business exports into a high-precision strategic roadmap, allowing for data-driven decisions that outpace traditional "gut-feeling" management. The SVI is a Plug-and-Play solution built for lean environments. It provides Fortune 500-level strategic clarity using nothing more than a standard business pivot table and a custom Python engine. It is designed to scale with the company, turning raw data into a continuous cycle of cash-flow optimization and growth.